

. Banach spaces Chapter 5.

. Recall Def: complete normed vector space $(V, \|\cdot\|)$.

. Examples: $\mathcal{C}(K; d_\infty)$ continuous functions from compact $K \rightarrow \mathbb{R}$ complete.

. $\mathcal{C}^k(K, \|\cdot\|_{k,\infty})$ space of continuous functions from $\mathbb{R}^n \supset K \rightarrow \mathbb{R}$ with derivatives up to order k bounded and continuous.

$$\|f\|_{k,\infty} = \sum_{j=0}^k \|f^{(j)}\|_\infty. \quad f^{(0)} \equiv f.$$

⚠ . $\mathcal{C}^\infty = \bigcap_{k \geq 0} \mathcal{C}^k$ is not a normed vector space

This is a Fréchet space (complete for metric based on semi-norms $\|f^{(j)}\|_\infty$ and given explicitly by

$$d(f, g) = \sum_{k=0}^{\infty} 2^{-k} \frac{\|f^{(k)} - g^{(k)}\|_\infty}{1 + \|f^{(k)} - g^{(k)}\|_\infty} \quad)$$

$$L^p(\Omega) = \{f: \Omega \rightarrow \mathbb{R} \text{ measurable} : \int_{\Omega} |f(x)|^p d\overset{\text{Lebesgue measure}}{\mu(x)} < \infty\}$$

- Examples :
 - L^p spaces , $W^{m,p}$ Sobolev spaces (up to m^{th} derivative in L^p).
 - $L^{\infty}(\Omega) = \{f: \Omega \rightarrow \mathbb{R} : \text{ess sup}_{x \in \Omega} |f(x)| < \infty\}$
 - $\ell_p(\mathbb{N})$ spaces of infinite sequences $x = (x_n)_{n \geq 1} ; x_n \in \mathbb{C}$.

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{\frac{1}{p}} \quad 1 \leq p < \infty ; \quad \|x\|_{\infty} = \sup_{n \geq 1} |x_n|$$

- Prop. : $(\ell_p(\mathbb{N}), \|\cdot\|_p)$ is Banach space for $1 \leq p \leq \infty$.

- $p = \infty$: same as continuous functions. Take $1 \leq p < \infty$.

- First: need to show $\|\cdot\|_p$ is a norm on what is obviously a vector space.

$$\|\lambda x\|_p = |\lambda| \|x\|_p \quad \checkmark \quad \|x\|_p = 0 \Rightarrow x_n = 0 \quad \forall n \Rightarrow x = 0 \quad \checkmark$$

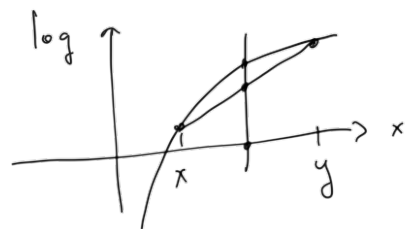
$$\|x + y\|_p \leq \|x\|_p + \|y\|_p \quad \text{called Minkowski inequality.}$$

$$\text{For } 1 \leq p < \infty, \text{ define } p' = \frac{p}{p-1} \quad 1 \leq p' < \infty \quad \frac{1}{p} + \frac{1}{p'} = 1.$$

Young's inequality $\left| \begin{array}{l} a, b \geq 0 \\ 1/p + 1/p' = 1 \end{array} \right| ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}$

when $p=p'=2$ this is
 $ab \leq \frac{1}{2}(a^2+b^2) \Leftrightarrow (a-b)^2 \geq 0$

• proof for $1 < p < \infty$ (clear when $p=1, p'=\infty$). $\log$ concave: $\log(\theta x + (1-\theta)y) \geq \theta \log x + (1-\theta) \log y$.



So: $\log\left(\frac{a^p}{p} + \frac{b^{p'}}{p'}\right) \geq \frac{1}{p} \log a^p + \frac{1}{p'} \log b^{p'} = \log ab$

$\theta = \frac{1}{p}, x = a^p, y = b^{p'}$. $\log \uparrow$ so $\frac{a^p}{p} + \frac{b^{p'}}{p'} \geq ab \quad \square$

Hölder's inequality $\left| \begin{array}{l} a_n \geq 0 \\ b_n \geq 0 \end{array} \right| \|ab\|_1 = \sum_{n \geq 1} a_n b_n \leq \left(\sum_{n \geq 1} a_n^p\right)^{\frac{1}{p}} \left(\sum_{n \geq 1} b_n^{p'}\right)^{\frac{1}{p'}} = \|a\|_p \|b\|_{p'}$

$\frac{1}{p} + \frac{1}{p'} = 1$

• proof: $\sum_{n \geq 1} a_n b_n = \sum_{n \geq 1} a_n \lambda^{\frac{1}{p}} b_n \lambda^{-\frac{1}{p}} \leq \sum_{n \geq 1} \frac{\lambda a_n^p}{p} + \frac{\lambda^{\frac{p'}{p}} b_n^{p'}}{p'} = \frac{\lambda}{p} \|a\|_p^p + \frac{\lambda^{\frac{1}{p'}}}{p'} \|b\|_{p'}^{p'}$

minimizing over λ gives $\lambda = \frac{\|b\|_{p'}^{p'}}{\|a\|_p^{p-1}}$. So $\|ab\|_1 \leq \underbrace{\left(\frac{1}{p} + \frac{1}{p'}\right)}_{\leq 1} \|a\|_p \|b\|_{p'} \quad \square$

• Both inequalities are important in their own right.

Note $ab = a \lambda^{\frac{1}{p}} \frac{b}{\lambda^{\frac{1}{p}}} \leq \frac{1}{p} \lambda^p a^p + \frac{1}{p'} \frac{1}{\lambda^{p'}} b^{p'}$ is also useful.

Minkowski $\|x+y\|_p \leq \|x\|_p + \|y\|_p$.

$$\|x+y\|_p^p = \sum_{n=1}^{\infty} |x_n + y_n|^{p-1} |x_n + y_n| \leq \sum_{n=1}^{\infty} |x_n + y_n|^{p-1} (|x_n| + |y_n|)$$

$$\stackrel{H.}{\leq} \left(\sum_{n=1}^{\infty} |x_n + y_n|^{\frac{p}{p-1}} \right)^{\frac{1}{p'}} \left\{ \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{\frac{1}{p}} + \left(\sum_{n=1}^{\infty} |y_n|^p \right)^{\frac{1}{p}} \right\}.$$

$$= \|x+y\|_p^{p-1} (\|x\|_p + \|y\|_p). \quad \text{If } \|x+y\|=0, \text{ clear; if not, as well. } \square$$

Thus, $(\ell_p, \|\cdot\|_p)$ n.v.s.p. Completeness?

proof. sequence $x^{(k)} \in \ell_p$ Cauchy: $\|x^{(k)} - x^{(l)}\|_p \rightarrow 0$ as $k, l \rightarrow \infty$.

(i) identify limit $|x_n^{(k)} - x_n^{(l)}|^p \rightarrow 0$ so $x_n^{(k)} - x_n^{(l)}$ Cauchy in $\mathbb{C}$

$\Rightarrow x_n^{(k)} \xrightarrow{k \rightarrow \infty} y_n$. Sequence $y = (y_n)_{n \geq 1}$ now defined in $\mathbb{C}^{\mathbb{N}}$.

(ii) $y \in \ell_p$? $\sum_{n=1}^N |y_n|^p = \sum_{n=1}^N \lim_{k \rightarrow \infty} |x_n^{(k)}|^p \leq \limsup_{k \rightarrow \infty} \|x^{(k)}\|_p^p \leq C.$

Indeed $x^{(k)}$ Cauchy $\Rightarrow x^{(k)}$ bounded uniformly.

Now $\sum_{n=1}^N |y_n|^p$ is non-decreasing and hence converges in $\mathbb{R} \cup \{+\infty\}$.

Above shows $\|y\|_p^p \leq C \Rightarrow y \in \ell_p$.

$$\begin{aligned}
 \text{(iii)} \quad x^{(k)} \xrightarrow{\ell_p} y ? \quad \|y - x^{(k)}\|_p^p &= \sum_{n=1}^{\infty} \lim_{\ell \rightarrow \infty} |x_n^{(\ell)} - x_n^{(k)}|^p \\
 &\leq \frac{\varepsilon}{2} + \sum_{n=1}^{N(\varepsilon)} \lim_{\ell \rightarrow \infty} |x_n^{(\ell)} - x_n^{(k)}|^p \leq \frac{\varepsilon}{2} + \lim_{\ell \rightarrow \infty} \sup_p \|x^{(\ell)} - x^{(k)}\|_p^p \leq \varepsilon \\
 &\quad \text{for } k \geq K(\varepsilon).
 \end{aligned}$$

Thus, $x^{(k)} \rightarrow y$ in $\ell_p \Rightarrow (\ell_p, \|\cdot\|_p)$ Banach space.

- $(L_p, \|\cdot\|_p)$ also Banach space with exactly the same proof (except that L_p is class of equivalence of functions ...)

• Bounded Linear Operators. $T: X \rightarrow Y$ normed vector spaces.

$$T \text{ linear} \Leftrightarrow T(\lambda f + \mu g) = \lambda T f + \mu T g, \quad \forall \lambda, \mu \in \mathbb{R} \text{ or } \mathbb{C}, \quad \forall f, g \in X.$$

- T one-to-one or injective when $Tx = Ty \Rightarrow x = y$
- T onto or surjective when $\forall z \in Y, \exists x \in X \text{ s.t. } Tx = z$.
- when T is one-to-one and onto (a.k.a. bijective), then we define $T^{-1}: Y \rightarrow X$.

Theorem. T^{-1} is linear.

- $T: X \rightarrow Y$ is bounded if there exists $M > 0$ s.t.

$$\|Tx\|_Y \leq M \|x\|_X \text{ for all } x \in X.$$

- Space of Linear operators is a vector space $(\lambda T + \mu \tilde{T})f = \lambda Tf + \mu \tilde{T}f$

- space of bounded linear operators is also a vector space.

- We now define a norm on space of bounded operators:

$$\begin{aligned} \|T\| &= \inf \{ M, \|Tx\| \leq M \|x\| \} \\ &= \sup_{x \neq 0} \frac{\|Tx\|_Y}{\|x\|_X} = \sup_{\|x\|_X = 1} \|Tx\|_Y. \end{aligned}$$

(Check this is a norm.)

- Thus, $\|Tx\|_Y \leq \|T\| \|x\|_X$. $\|\cdot\|$ is called uniform norm or operator norm.

- This space is even an algebra: $T: X \rightarrow Y$ and $S: Y \rightarrow Z$ bounded
 $\Rightarrow ST: X \rightarrow Z$ bounded.

• Examples . $X = (C([0,1]), \|\cdot\|_\infty)$ $T: \begin{array}{ccc} X & \xrightarrow{\quad} & X \\ f & \xrightarrow{T} & \int_0^x f(t) dt \end{array}$

$$\|Tf\|_\infty = \sup_{x \in [0,1]} \left| \int_0^x f(t) dt \right| \leq \sup_{x \in [0,1]} \int_0^x \|f\|_\infty dt \leq \sup_{x \in [0,1]} \int_0^x 1 dt \|f\|_\infty = \|f\|_\infty.$$

So, $\|T\| \leq 1$.

but $1 \rightarrow T1 = \int_0^x 1 dt = x$ and $\|x\|_\infty = 1$. So $\|T\| \geq 1$

$\|T\| = 1$.

• Taking derivatives is "unbounded". Take $C^\infty([0,1]) = \bigcap_{k \geq 0} C^k([0,1])$

Then $X = (C^\infty([0,1]), \|\cdot\|_\infty)$ is n.v.s.p (though not complete).

$T: \begin{array}{ccc} X & \rightarrow & X \\ f & \rightarrow & f' \end{array}$. This is unbounded

with $\|f_n\|_\infty = e^n$ and $\|f'_n\|_\infty = n e^n$

$$\frac{\|T f_n\|}{\|f_n\|} = n \text{ unbounded.}$$

- Taking derivatives is bounded.

$$\begin{cases} X = (C^1[0,1], \|\cdot\|_{1,\infty}) \\ Y = (C^0[0,1], \|\cdot\|_\infty) \end{cases}$$
 $\|f\|_{1,\infty} = \|f\|_\infty + \|f'\|_\infty$
 $\|Tf\|_\infty = \|f'\|_\infty \leq \|f\|_{1,\infty}$

Then $T: X \rightarrow Y$
 $f \mapsto f'$ is bounded with $\|T\| \leq 1$.

- So, topologies on X and Y matter to decide whether T is bounded!

- $A: X \rightarrow Y$ $\dim X = n$, $\dim Y = m$. $|X \overset{\text{isomorphic}}{\simeq} \mathbb{C}^n; Y \simeq \mathbb{C}^m$
 (or $\mathbb{R} \dots$)

Then A can be represented by a matrix $\underline{\underline{A}} = (a_{ij})_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$

$\begin{matrix} \xrightarrow{n} \\ \boxed{A} \\ \xleftarrow{m} \end{matrix}$, Exercise: check construction

any linear mapping can be written as matrix form

• Theorem. T linear. Then T bounded $\Leftrightarrow T$ continuous $\Leftrightarrow T$ continuous at 0.

• proof: T bounded $\Rightarrow \|Tx - Ty\|_Y = \|T(x-y)\|_Y \leq \|T\| \|x-y\|$

So T Lipschitz with $\text{Lip}(T) = \|T\|$; $\delta = \varepsilon / \|T\|$.

• T continuous at 0 $\forall \varepsilon, \exists \delta, \|x\|_X \leq \delta \Rightarrow \|Tx\|_Y \leq \varepsilon$

$TO = 0$

• take $\varepsilon = 1$ and $\delta = \delta(1)$ so that $\|x\|_X \leq \delta \Rightarrow \|Tx\|_Y \leq 1$

• this means $T(B_\delta(0)) \subset B_1(0)$. By linearity $T(B_{\frac{1}{\delta}}(0)) \subset B_1(0)$

take $\lambda = \frac{1}{\delta} \Rightarrow T(B_1(0)) \subset B_{\frac{1}{\delta}}(0)$. But this means $\|Tx\| \leq \frac{1}{\delta}$ when $\|x\| \leq 1$.

• Seen another way: Take $0 \neq z \in X$. $\|\frac{\delta z}{\|z\|_X}\|_X \leq \delta \Rightarrow \|T \frac{\delta z}{\|z\|_X}\|_Y \leq 1$

$\Rightarrow \|Tz\|_Y \leq \frac{1}{\delta} \|z\|_X$ by linearity. Thus, $\|T\| \leq 1/\delta$ bounded $\square$

• Theorem Bounded Linear Transformation (BLT). $T: M \subset X \text{ n.v.s.p.} \rightarrow Y$ Banach.

$$\|T\| = \sup \frac{\|Tz\|}{\|z\|}$$

M dense in X i.e. $\bar{M} = X$. Then T bounded $\Rightarrow$ there is a unique extension

$\bar{T}: X \rightarrow Y$ s.t. $\bar{T}x = Tx$ for $x \in M$ and $\|\bar{T}\| = \|T\|$.

• Sketch of proof: $M \ni x_n \rightarrow x \in X$. $y_n = Tx_n$. T bounded implies

$$\|y_n - y_m\|_Y = \|T(x_n - x_m)\| \leq \|T\| \|x_n - x_m\| \rightarrow 0 \text{ so } y_n \text{ Cauchy.}$$

$y_n \rightarrow y =: \tilde{T}x$. Check $\tilde{T}$ extension with $\|\tilde{T}\| = \|T\|$ and uniqueness. $\square$

Thm. Open mapping theorem.

$T: X \rightarrow Y$ one-to-one, onto, bounded, linear map; X, Y Banach.

← inverse version

Then $T^{-1}: Y \rightarrow X$ is bounded.

• No control over $\|T^{-1}\|$. Proof not constructive.

• Other formulation: $T: X \rightarrow Y$ onto, bounded, linear; X, Y Banach.

← open mapping version

(open mapping thm.) Then $\exists c > 0$ s.t. $T(B_X(0,1)) \supset B_Y(0,c)$.

• This means T (open) is open and hence the name of the theorem.

• Remember that continuity $\Leftrightarrow T^{-1}$ (open) is open.

• proof from other formulation: $\exists c$ s.t. $\|Tx\|_Y \leq c \Rightarrow \|x\|_X \leq 1$.

Take $Tx \neq 0$, define $\lambda = \frac{c}{\|Tx\|}$. Then $\|T\lambda x\| = c$ so that $\|\lambda x\| \leq 1$

which means $\|x\| \leq \frac{1}{c} \|Tx\|$. Since T bijjective, $T^{-1}: Y \rightarrow X$ is defined

and $y = Tx \Leftrightarrow x = T^{-1}y$. So, $\|T^{-1}y\| \leq \frac{1}{c} \|y\|$; T^{-1} bounded $\square$

Thm. then also called bounded inverse theorem.

Corollary: X Banach space for $\|\cdot\|_1$ and $\|\cdot\|_2$. Assume $\|\cdot\|_1 \leq C \|\cdot\|_2$

Then there is C' s.t. $\|\cdot\|_2 \leq C' \|\cdot\|_1$. $\|\cdot\|_1 \sim \|\cdot\|_2$

proof: $(X, \|\cdot\|_2) \xrightarrow{\text{Id}} (X, \|\cdot\|_1)$ is bounded linear, one-to-one, onto.

So, Open mapping $\Rightarrow$ $\text{Id}: (X, \|\cdot\|_2) \rightarrow (X, \|\cdot\|_1)$ is bounded $\square$

Corollary: Assume $(X, \|\cdot\|_1)$ and $(X, \|\cdot\|_2)$ are Banach spaces with Same Cauchy limits. Then $\|\cdot\|_1 \sim \|\cdot\|_2$.

proof. We show that $(X, \|\cdot\|_1 + \|\cdot\|_2)$ is Banach. Indeed, y_n Cauchy for $\|\cdot\|_1 + \|\cdot\|_2 \Rightarrow$ Cauchy for $\|\cdot\|_1$ and $\|\cdot\|_2$. But limits are equal, so Cauchy for sum.

Now $\|\cdot\|_1 \leq \|\cdot\|_1 + \|\cdot\|_2$. Above $\nearrow$ shows $\|\cdot\|_2 + \|\cdot\|_1 \leq C \|\cdot\|_1$.

Similarly $\|\cdot\|_1 \leq C' \|\cdot\|_2$. corollary

N.B. Two B. spaces $(X, \|\cdot\|_1)$ and $(X, \|\cdot\|_2)$ with different limits of Cauchy sequences may exist (abstract constructions).

Closed graph theorem. $T: X \rightarrow Y$ linear. X, Y Banach spaces.

$$\text{Graph}(T) = \bigcup_{x \in X} [x, Tx] \subset X \times Y$$

Def: $T: X \rightarrow Y$ linear is said to be closed if Graph(T) is closed.

This means $\{(x_n, Tx_n) \rightarrow (x, y)\} \Rightarrow \{y = Tx\}$.

Theorem: T closed $\Rightarrow T$ bounded.

proof: $(X, \underbrace{\|\cdot\|_X + \|T\cdot\|_Y}_{\text{graph norm}})$. Show that T closed $\Rightarrow (X, \|\cdot\|_X + \|T\cdot\|_Y)$ Banach.

Thus, from previous corollary $\|\cdot\|_X + \|T\cdot\|_Y \leq C\|\cdot\|_X \Rightarrow T$ bounded.

N.B. T bounded $\Rightarrow T$ continuous $\Rightarrow T$ closed.

Sometimes useful to show that "abstractly" defined operators are bounded.

inverse hold if linear and Banach.

• Kernel, range of linear operators.

Def $T: X \rightarrow Y$ linear $\text{Ker}(T) \equiv N(T) = \{ x \in X, Tx = 0 \}$

$$\text{Ran}(T) \equiv R(T) = \{ y \in Y \text{ s.t. } \exists x \in X \text{ s.t. } y = Tx \} \\ = \bigcup_{x \in X} Tx.$$

Theorem $\text{Ker}(T)$ and $\text{Ran}(T)$ are linear subspaces of X and Y , respectively.

• X, Y nvsps T bounded. Then $\text{Ker}(T)$ is closed. $\{x_n \rightarrow x; Tx_n = 0 \Rightarrow Tx = 0\}$

• $T: X \rightarrow Y$ one-to-one (injective) iff $\text{Ker}(T) = \{0\}$.

onto (surjective) iff $\text{Ran}(T) = Y$.

• $\dim \text{Ker}(T)$ is called nullity of T ; $\dim \text{Ran}(T)$ is called rank of T .

• Examples $X = (C([0,1]), \|\cdot\|_\infty)$. $K: X \rightarrow X$ $f \mapsto Kf(x) = \int_0^x f(t) dt$. $\|K\| = 1$.

• $\text{Ran}(K) = \{ f \in C^1([0,1]) \text{ s.t. } f(0) = 0 \}$. $\overline{\text{Ran}(K)}^{\|\cdot\|_\infty} = \{ f \in C^0([0,1]); f(0) = 0 \} \neq \text{Ran}(K)$.

• $T = I + K: X \rightarrow X$ $Tf(x) = f(x) + \int_0^x f(t) dt$.

• Check: $Tf = g \Leftrightarrow f(x) = g(x) - \int_0^x e^{-(x-y)} g(y) dy =: T^{-1}g(x)$.

• $\text{Ran}(T) = X$ (closed). 13

Proposition $T: X \rightarrow Y$ bounded X, Y Banach

Then: $\{ \exists c > 0 \text{ s.t. } c\|x\| \leq \|Tx\|, \forall x \in X \} \Leftrightarrow \{ \text{Ran}(T) \text{ closed and } \text{Ker}(T) = \{0\} \}$.

Method of a priori estimates: If we can show Tx bounded below, then $Tx = y$ has unique solution!

proof $\Rightarrow$. $\text{Ker } T = \{0\}$ since $Tx = 0 \Rightarrow x = 0$. Let $y_n \in \text{Ran}(T)$, $y_n \rightarrow y$. Do we have $y \in \text{Ran}(T)$?

$y_n = Tx_n$ $\|x_n - x_m\| \leq \frac{1}{c} \|Tx_n - Tx_m\| = \frac{1}{c} \|y_n - y_m\| \rightarrow 0 \Rightarrow$ Cauchy $x_n \rightarrow x \in X$.

By continuity $y_n = Tx_n \rightarrow Tx = y$ $\square$ $\{ Y \text{ Banach not necessary for a priori estimates} \}$.

$\Leftarrow$ $T: X \rightarrow \text{Ran}(T)$ is one-to-one and onto. So $T^{-1}: \text{Ran}(T) \rightarrow X$ is defined.

$\text{Ran}(T)$ closed $\Rightarrow$ $\text{Ran}(T)$ Banach space for $\|\cdot\|_Y$. Open mapping $\Rightarrow T^{-1}$ bounded.

$\|T^{-1}\| \leq \frac{1}{c}$ is equivalent to $c\|x\| \leq \|Tx\| \forall x \in X \square$

• Finite dimensional spaces. Read H.N. Results are:

• $(X, \|\cdot\|)$ dimension ^{finite} n . Then $\|\cdot\| \sim \|\cdot\|_1$ $\|x\|_1 = \sum_{j=1}^n |x_j|$.

• Every finite-dim. n.v.sp. is Banach.

• Every finite-dim. subspace of (arbitrary) n.v.sp. is closed.

• Every linear operator on finite dim space (aka matrix) is bounded.

• $(X, \|\cdot\|_1)$ and $(X, \|\cdot\|_2)$ on finite dim. space $\Rightarrow \|\cdot\|_1 \sim \|\cdot\|_2$.

• Bounded operators $\mathcal{B}(X, Y) \equiv \mathcal{L}(X, Y)$ ^{on finite} vector space

• $(\mathcal{B}(X, Y), \|\cdot\|)$ with $\|T\| = \sup_{\|x\|=1} \|Tx\|$ (operator norm / uniform norm) is n.v.sp.

• Thm $T \in \mathcal{B}(X, Y), S \in \mathcal{B}(Y, Z) \Rightarrow ST \in \mathcal{B}(X, Z)$ with $\|ST\| \leq \|S\| \|T\|$.

• Definition Commutator $[S, T] = ST - TS$ not necessarily vanishing!

15 • $[S, T] = 0$: S and T commute.

Def Convergence. $T_n \rightarrow T$ uniformly (in operator norm) if $\|T_n - T\| \rightarrow 0$ as $n \rightarrow \infty$.

Theorem X n.v.s.p and Y Banach. Then $(\mathcal{B}(X, Y), \|\cdot\|)$ is Banach.

Proof: (i) T_n Cauchy $\|T_n - T_m\| \rightarrow 0 \Rightarrow \|T_n x - T_m x\|_Y \rightarrow 0$ for $x \in X$ fixed.

Thus, $T_n x$ Cauchy in $Y \Rightarrow T_n x \rightarrow y$ in Y . Define T as $x \mapsto T x = y$.

Check that T is linear.

(ii) Above convergence $\Rightarrow \forall \varepsilon > 0, \exists M = M(x, \varepsilon), \forall m \geq M, \|T x - T_m x\| \leq \frac{\varepsilon}{2} \|x\|$.

so $\|T_n x - T x\| \leq \underbrace{\|T_n x - T_m x\|}_{\leq \|T_n - T_m\| \|x\|} + \|T_m x - T x\| \leq \varepsilon \|x\|$ in large $\because n \geq N(\varepsilon)$ large.

(iii) This shows $\|T x\| \leq \|T_n x - T x\| + \|T_n x\| \leq C \|x\| \Rightarrow T$ bounded

$\|T_n x - T x\| \leq \varepsilon \|x\|$ for n large $\Rightarrow T_n \rightarrow T$ uniformly $\square$
